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Abstract 摘要

In the field of autonomous driving or robotics, simultaneous localization and mapping (SLAM) and multi-object tracking (MOT) are two fundamental problems and are generally applied separately. Solutions to SLAM and MOT usually rely on certain assumptions, such as the static environment assumption for SLAM and the accurate ego-vehicle pose assumption for MOT. But in complex dynamic environments, it is difficult or even impossible to meet these assumptions. Therefore, the SLAMMOT, i.e., simultaneous localization, mapping, and moving object tracking, integrated system of SLAM and object tracking, has emerged for autonomous vehicles in dynamic environments. However, many conventional SLAMMOT solutions directly perform data association on the predictions and detections for object tracking, but ignore their quality. In practice, inaccurate predictions caused by continuous multi-frame missed detections in temporary occlusion scenarios, may degrade the performance of tracking, thereby affecting SLAMMOT. To address this challenge, this paper presents a LiDAR SLAMMOT based on confidence-guided data association (Conf SLAMMOT) method, which tightly couples the LiDAR SLAM and the confidence-guided data association based multi-object tracking into a graph optimization backend for estimating the state of the ego-vehicle and objects simultaneously. The confidence of prediction and detection are applied in the factor graph-based multi-object tracking for its data association, which not only avoids the performance degradation caused by incorrect initial assignments in some filter-based methods but also handles issues such as continuous missed detection in tracking while also improving the overall performance of SLAMMOT. Various comparative experiments demonstrate the superior advantages of Conf SLAMMOT, especially in scenes with some missed detections.

在自動駕駛或機器人領域中，同步定位與繪圖 (SLAM) 和多目標追蹤 (MOT) 是兩個基本問題，通常會分開應用。SLAM 和 MOT 的解決方案通常依賴於某些假設，例如 SLAM 的靜態環境假設和 MOT 的精確自我車輛姿勢假設。但在複雜的動態環境中，要滿足這些假設是很困難的，甚至是不可能的。因此，針對動態環境下的自主車輛，出現了 SLAMMOT，即同時定位、繪圖與移動物體追蹤，是 SLAM 與物體追蹤的整合系統。然而，許多傳統的 SLAMMOT 解決方案直接對物體追蹤的預測與偵測執行資料聯結，卻忽略了其品質。實際上，在臨時遮蔽的情況下，由於連續的多幀遺漏偵測所造成的不準確預測，可能會降低追蹤的效能，進而影響 SLAMMOT。針對此一挑戰，本論文提出一種基於信心導向資料關聯的 LiDAR SLAMMOT 方法 (Conf SLAMMOT)，該方法將 LiDAR SLAM 與基於信心導向資料關聯的多物體追蹤緊密結合到一個圖形最佳化後端，以同時估計自我車輛與物體的狀態。預測與偵測的信賴度應用在因子圖基於多物體追蹤的資料關聯中，不僅避免了一些基於濾波的方法中因初始分配錯誤而造成的性能下降，也處理了追蹤中連續漏檢等問題，同時也提高了 SLAMMOT 的整體性能。各種比較實驗證明了 Conf SLAMMOT 的優越性，尤其是在有一些遺漏偵測的場景中。

Index Terms-simultaneous localization, map, and moving object tracking (SLAMMOT), LiDAR SLAMMOT, multi-object tracking (MOT), confidence-guided data association

索引標題-同步定位、地圖與移動物件追蹤 (SLAMMOT)、LiDAR SLAMMOT、多物件追蹤 (MOT)、信心導向資料關聯

I. Introduction I.簡介

SIMULTANEOUS localization and mapping (SLAM) and multi-object tracking (MOT) are two fundamental problems for autonomous driving or robotic systems. SLAM aims to simultaneously model the surrounding static environment and accurately estimate the vehicle pose within the constructed map. MOT focuses on perceiving and estimating the states of surrounding moving objects. Currently, solutions for SLAM and MOT are usually applied separately and rely on certain assumptions. Most SLAM solutions rely on the assumption of a static environment. Many mature SLAM solutions have been proposed for single-vehicle applications [1]-[3],

but they still face significant challenges in dynamic environments

同步定位與繪圖 (SLAM) 和多目標追蹤 (MOT) 是自動駕駛或機器人系統的兩個基本問題。SLAM 旨在同時對周遭靜態環境建模，並在建構的地圖中精確估計車輛姿勢。MOT 著重於感知和估計周遭移動物件的狀態。目前，SLAM 和 MOT 的解決方案通常分開應用，並依賴特定的假設。大多數 SLAM 解決方案都依賴於靜態環境的假設。許多成熟的 SLAM 解決方案已針對單車應用提出 [1]-[3]，但在動態環境中仍面臨重大挑戰。

containing a large number of moving objects, leading to degraded performance or even failure. On the other hand, multiobject tracking commonly assumes the accurate ego-vehicle pose as a known prior, placing little focus on its estimation. This dependence means that traditional MOT methods are significantly reliant on the precision of ego-pose estimation. However, in unknown dynamic environments, it is difficult to ensure the presence of reliable static structures, which often leads to an unreliable pose estimation.

包含大量移動物件，導致效能降低甚至失敗。另一方面，多目標追蹤通常會假設精確的自我車輛姿勢為已知先驗，而不太注重其估計。這種依賴性意味著傳統的 MOT 方法嚴重依賴於自我姿勢估計的精確度。然而，在未知的動態環境中，很難確保存在可靠的靜態結構，這往往會導致不可靠的姿勢估計。

Therefore, coupled SLAM and multi-object tracking can be a viable approach due to the inherent interdependence between the two challenges mentioned. Simultaneous localization, mapping, and moving object tracking (SLAMMOT), which was firstly presented in [4], coupling SLAM and MOT, allows for state estimation of ego-vehicle and moving objects simultaneously in dynamic environments. SLAMMOT can not only guarantee the performance of SLAM or MOT when the assumptions they rely on are difficult or even impossible to hold in complex dynamic environments, but also can mutually benefit each other. Specifically, for SLAM, considering the impact of moving objects transforms them from mere dynamic interference into auxiliary information that enhances pose estimation. And accurate ego-pose estimation from SLAM can significantly improve the estimation of object states for MOT.

因此，由於上述兩項挑戰之間固有的相互依存性，耦合 SLAM 與多目標追蹤可以是一種可行的方法。同時定位、映射與移動物體追蹤 (Simultaneous localization, mapping, and moving object tracking · SLAMMOT) 最早在 [4] 中提出，結合 SLAM 與 MOT，可在動態環境中同時進行自我車輛與移動物體的狀態估計。當 SLAM 或 MOT 所依賴的假設在複雜的動態環境中難以成立甚至不可能成立時，SLAMMOT 不僅可以保證 SLAM 或 MOT 的效能，而且可以互惠互利。具體而言，對 SLAM 而言，考慮到移動物體的影響，可將其從純粹的動態干擾轉換為可增強姿態估計的輔助資訊。而由 SLAM 所得到的精確自我姿勢估計，可大幅改善 MOT 的物件狀態估計。

In recent years, there has been significant progress in the research of SLAMMOT technology, such as vision-based SLAMMOT methods [5]-[7] and LiDAR-based methods [8][10]. While vision-based methods play an important role in the robotics and intelligent vehicle field, the advantages of LiDAR, such as adaptability to all kinds of weather conditions, high precision in distance measurements, high ranging accuracy, and wide horizontal field of view, have attracted more attention of researchers. However, there are still challenges in existing LiDAR-based SLAMMOT solutions. Most existing SLAMMOT methods directly adopt some kind of association strategy for data association between object prediction results and detections. However, object predictions are not always accurate, especially for objects that are missed in consecutive frames due to temporary occluded or distant objects, whose prediction errors may accumulate without receiving state updates. At the same time, some low-quality detections can also affect the results of data association. The inaccurate prediction and lowquality detection can destabilize the state of the matched track, or eventually lead to track failure and degrade the performance of SLAMMOT simultaneously. This tracking issue could pose a threat to driving safety if severe. Therefore, the confidence of predictions and detections should be considered during the data association in tracking of SLAMMOT.

近年來，SLAMMOT 技術的研究有了顯著的進展，例如基於視覺的 SLAMMOT 方法[5]-[7]和基於 LiDAR 的方法[8][10]。以視覺為基礎的 SLAMMOT 方法在機器人與智慧車輛領域中扮演著重要的角色，而 LiDAR 的優點，例如可適應各種天候條件、高精確度的距離量測、高測距精度以及寬廣的水平視場等，則吸引了更多研究者的注意。然而，現有以 LiDAR 為基礎的 SLAMMOT 解決方案仍存在許多挑戰。現有的 SLAMMOT 方法大多直接採用某種關聯策略，將物體預測結果與偵測結果進行資料關聯。然而，物體預測並不總是精確的，尤其是對於因暫時遮蔽或距離較遠而在連續幀中遺漏的物體，其預測誤差可能會在未收到狀態更新的情況下累積。同時，一些低品質的偵測也會影響資料關聯的結果。不準確的預測與低品質檢測可能會使匹配軌跡的狀態不穩定，或最終導致軌跡失效，同時降低 SLAMMOT 的效能。這種追蹤問題嚴重時可能會對行車安全構成威脅。因此，在 SLAMMOT 追蹤的資料關聯過程中，應考慮預測與偵測的信賴度。

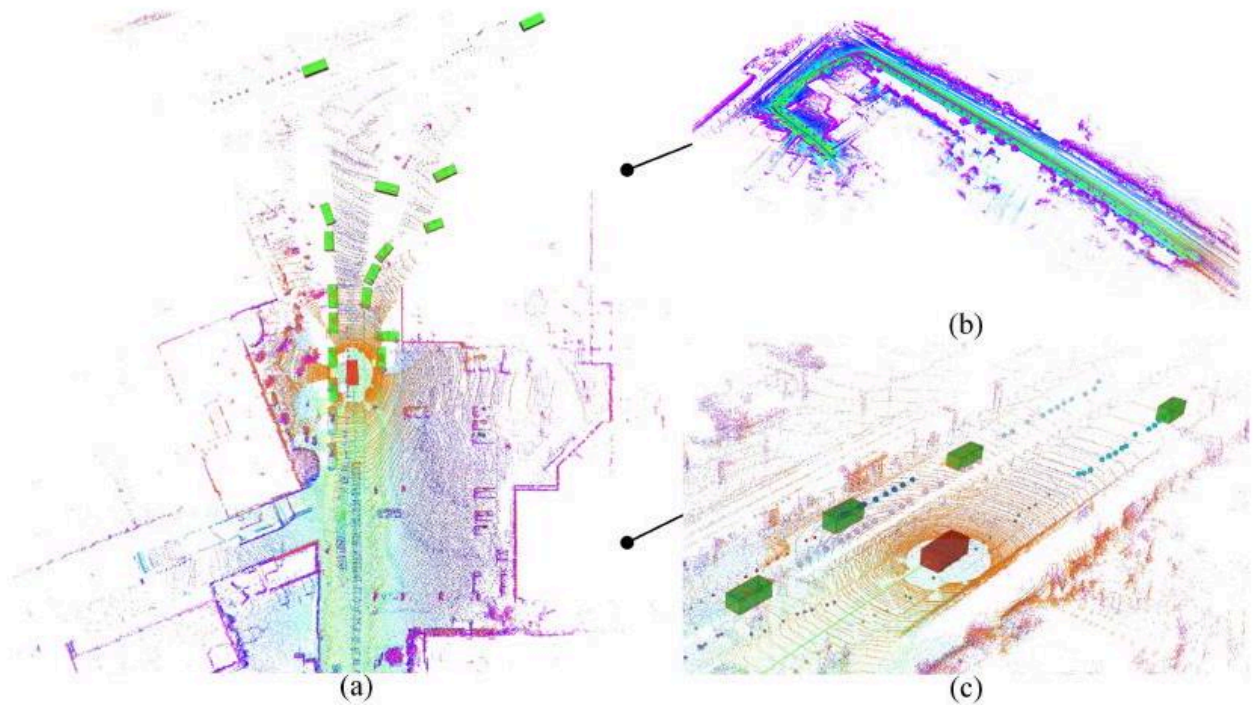


Fig. 1. Results of proposed Conf SLAMMOT solution in KITTI Tracking dataset [11]. (b) and © denote the generated complete point cloud map and details of the result shown in (a), respectively. Red and green fully covered bounding boxes denote the ego-vehicle and tracked objects, and red rectangular border denotes detected objects. The green solid line and the dotted lines represent the trajectories of ego-vehicle and tracked objects, respectively.

圖 1. 建議的 Conf SLAMMOT 解決方案在 KITTI 追蹤資料集 [11] 中的結果。(b) 和 © 分別表示產生的完整點雲圖和 (a) 所示結果的詳細資訊。紅色和綠色完全覆蓋的邊框表示自我車輛與追蹤到的物件，紅色矩形邊框表示偵測到的物件。綠色實線與虛線分別代表自我車輛與追蹤物件的軌跡。

In this paper, we present the Conf SLAMMOT, a tightly coupled LiDAR-based SLAMMOT solution that performs confidence-guided data association in multi-object tracking, which can not only estimate the state of the ego-vehicle and moving objects simultaneously in dynamic environments, but also overcome the continuous missed detections caused by temporary occlusion or objects being far away in tracking. Fig. 1 illustrates the results of the proposed Conf SLAMMOT solution in the KITTI Tracking dataset [11]. The Conf SLAMMOT includes a SLAM module, an object detection module, a confidence-guided data association based multi-object tracking relying on a factor graph, and a joint graph optimization backend for coupling and tracking. Herein, the SLAM and object detection modules are alternative and can be any representative solutions in the corresponding fields. The factor graph-based tracking utilizes the confidence of prediction and detection for its implicit data association, which can not only avoid the performance degradation caused by incorrect initial assignments, which is common in some filter-based methods, but also handle issues such as continuous missed detection in tracking. The partial factor graphs provided separately by the SLAM and object tracking modules are integrated into the graph optimization backend of Conf SLAMMOT, and the states of the ego-vehicle and the objects are jointly optimized. And the proposed method has achieved superior results in both SLAM and multi-object tracking tasks in various scenarios. The main highlights of the work are as follows:

在本文中，我們提出 Conf SLAMMOT，這是一種緊密耦合 LiDAR 的 SLAMMOT 解決方案，可在多目標追蹤中執行置信度導向的資料關聯，不僅能在動態環境中同時估計自我車輛與移動物體的狀態，也能克服在追蹤中因臨時遮擋或物體距離太遠而造成的連續漏測。圖 1 展示所提出的 Conf SLAMMOT 解決方案在 KITTI Tracking 資料集 [11] 中的結果。Conf SLAMMOT 包含一個 SLAM 模組、一個物體偵測模組、一個基於因子圖的信心導向資料關聯多目標追蹤，以及一個用於耦合與追蹤的聯合圖最佳化後端。在此，SLAM 模組與物體偵測模組是可選擇的，可以是相對應領域中任何具有代表性的解決方案。以因數圖為基礎的追蹤，利用預測與偵測的置信度進行隱含的資料關聯，不僅可以避免某些以濾波為基礎的方法中常見的初始分配錯誤所造成的效能下降，也可以處理追蹤中連續遺漏偵測等問題。將 SLAM 模組與物體追蹤模組分別提供的部分因子圖，整合至 Conf SLAMMOT 的圖最佳化後端，共同最佳化自我車輛與物體的狀態。而所提出的方法在 SLAM 與多物體追蹤任務的各種情境下都取得了優異的結果。本工作的主要重點如下：

We present a LiDAR based SLAMMOT method that tightly couples SLAM and confidence-guided data association based multi-object tracking, allowing simultaneous estimation of the ego-vehicle and moving

objects states, and achieving reliable tracking, especially in missed detection scenarios.

我們提出一種以 LiDAR 為基礎的 SLAMMOT 方法，該方法緊密結合了 SLAM 與以信賴導向資料關聯為基礎的多目標追蹤，允許同時估計自我車輛與移動物體的狀態，並實現可靠的追蹤，尤其是在遺漏偵測的情況下。

We perform a predictor that applies a constant turn rate and velocity model to estimate the future state of object, and utilizes a prediction confidence model and the confidence score of detection to guide data association by flexibly adjusting the implicit search range of data association.

我們執行一個預測器，應用恆定的轉向率和速度模型來估計物體的未來狀態，並利用預測置信度模型和偵測的置信度分數，透過彈性調整資料關聯的隱含搜尋範圍來引導資料關聯。

The factor graph based tracking utilizes the confidence of prediction and detection for its data association, avoiding issues from wrong initial assignments in some filter-based methods and effectively recovering tracking of continuously missed detections due to objects being occluded or distant.

基於因子圖的追蹤利用預測與偵測的置信度來進行資料關聯，避免了某些基於篩選器的方法中初始分配錯誤的問題，並能有效恢復因物件被遮蔽或距離較遠而持續遺漏偵測的追蹤。

II. RELated WORK II.相關工作

A. LiDAR based SLAM

A.基於 LiDAR 的 SLAM

In the field of LiDAR-based SLAM, many solutions have been proposed. One of the most popular methods is LOAM [12], which is a low-drift and real-time LiDAR odometry and mapping method. However, the performance of LOAM is limited when resources are scarce or when operating in noisy environments. To address these limitations, LeGO-LOAM [1] has been proposed, which is a lightweight and groundoptimized LOAM method and offers improved performance with limited resources. Additionally, auxiliary sensors like IMU can be closely integrated with LiDAR to improve SLAM performance. Researchers presents LIOM [13], which uses graph-based optimization to refine the extrinsic parameters of LiDAR and IMU, as well as IMU bias and robot pose. Following this, the LIO-SAM method [2] was introduced, estimating IMU bias within a factor graph framework and providing better real-time performance and more accurate state estimation compared to LIOM. To simplify computations, FAST-LIO [14] was developed, employing a tightly-coupled LiDAR-inertial odometry method based on an iterated Kalman filter with a new Kalman gain. Researchers then enhanced FAST-LIO with FAST-LIO2 [3], which directly matches raw point clouds without feature extraction and boosts mapping velocity and odometry accuracy using iKD-Tree. Nevertheless, all of these LiDAR-based methods rely on the static environment assumption and are easily affected by dynamic objects.

在以 LiDAR 為基礎的 SLAM 領域中，已有許多解決方案被提出。其中最受歡迎的方法是 LOAM [12]，這是一種低漂移且即時的 LiDAR 測距與繪圖方法。然而，當資源稀少或在嘈雜環境下操作時，LOAM 的效能就會受到限制。為了解決這些限制，有人提出 LeGO-LOAM [1]，這是一種輕量且經過地面優化的 LOAM 方法，可在資源有限的情況下提供最佳的效能。此外，IMU 等輔助感測器可與 LiDAR 緊密整合，以改善 SLAM 效能。研究人員提出了 LIOM [13]，該方法使用圖形最佳化來精細 LiDAR 與 IMU 的外在參數，以及 IMU 偏差與機器人姿勢。之後，又提出了 LIO-SAM 方法 [2]，在因子圖框架內估算 IMU 偏差，與 LIOM 相比，提供了更好的即時效能與更精確的狀態估算。為了簡化計算，研究人員開發了 FAST-LIO [14]，採用緊密耦合的 LiDAR 慣性里程測量方法，該方法基於具有新 Kalman 增益的迭代 Kalman 篩選器。研究人員隨後使用 FAST-LIO2 [3] 增強了 FAST-LIO，該方法不需進行特徵萃取即可直接匹配原始點雲，並使用 iKD-Tree 提升了測圖速度和里程計準確度。然而，所有這些以 LiDAR 為基礎的方法都依賴於靜態環境假設，很容易受到動態物件的影響。

For LiDAR-based SLAM in dynamic environments, researchers have also made some efforts. In [15] and [16], the researchers use convolutional neural networks (CNN) to detect moving objects from LiDAR point clouds, and simply filter all points belonging to dynamic objects for reliable SLAM in dynamic environments. In [17], a two-stream CNN method was applied to segment potential dynamic objects from point clouds. After removing the points associated with the potential dynamic objects, the point clouds were used to perform SLAM. In [18], researchers present the RF-LIO, which detects moving points in a LiDAR scan using adaptive multi-resolution range images and removes them, then realizes a tightly-coupled LiDAR inertial odometry in high dynamic environments. However, these methods may overlook valuable information, like points from stationary objects

(e.g., parked vehicles), or face tracking errors due to the absence of a relationship between the ego-vehicle state and object state.

對於動態環境中基於 LiDAR 的 SLAM，研究人員也做了一些努力。在[15]與[16]中，研究人員使用卷積神經網路 (CNN) 從 LiDAR 點雲中偵測移動物體，並簡單過濾所有屬於動態物體的點，以達到動態環境中可靠的 SLAM。在 [17] 中，應用雙流 CNN 方法來分割點雲中的潛在動態物件。在移除與潛在動態物件相關的點之後，點雲被用來執行 SLAM。在 [18] 中，研究人員提出 RF-LIO，利用自適應多解析度範圍影像偵測 LiDAR 掃描中的移動點並將其移除，然後在高動態環境中實現緊密耦合的 LiDAR 慣性里程計。然而，這些方法可能會忽略一些有價值的資訊，例如來自靜止物體 (例如停放的車輛) 的點，或是因為自我車輛狀態與物體狀態之間缺乏關係而造成的臉部追蹤錯誤。

B. Multi-object Tracking

B.多目標追蹤

Multi-object tracking is a crucial module of the vehicle perception system. Existing tracking methods can be categorized into two groups: non-detection based methods and detection based tracking. In the non-detection based method, one approach is to integrate detection and tracking into a single network, as demonstrated by JDE [19] and CenterTrack [20]. Another approach is attention mechanism based tracking, exemplified by TrackFormer [21]. However, these strategies suffer from drawbacks such as poor interpretability and generalization, and are primarily used in image based

多目標追蹤是車輛感知系統的重要模組。現有的追蹤方法可分為兩類：基於非偵測的方法和基於偵測的追蹤方法。在非偵測型方法中，一種方法是將偵測與追蹤整合為單一網路，如 JDE [19] 與 CenterTrack [20] 所示。另一種方法是基於注意機制的追蹤，以 TrackFormer [21] 為例。然而，這些策略都有一些缺點，例如可解讀性和概括性較差，而且主要用於以影像為基礎的

object tracking. Therefore, LiDAR based multi-object tracking typically employs a detection based method.

物件追蹤。因此，以 LiDAR 為基礎的多目標追蹤通常會採用以偵測為基礎的方法。

In [22], researchers proposed AB3DMOT, which is based on a baseline for 3D multi-object tracking, featuring a lightweight architecture and low computational demands, enabling realtime operation. In [23], researchers introduced a new 3D multi-object tracker guided by prediction confidence for data association, which can better leverage object features for multi-object tracking in point clouds. These two methods are both advanced LiDAR-only real-time 3D multi-object tracking techniques, and they utilize detection results obtained from LiDAR point cloud data and employ Kalman filtering with Hungarian matching and greedy algorithm respectively for state estimation and data association. However, if the initial assignment in data association is incorrect, such filterbased approaches often struggle, which can easily occur. In [24], researchers present a factor graph-based 3D multi-object tracking (FG-3DMOT) method, which represents the detection results as a Gaussian mixture model (GMM) and is integrated into a factor graph framework. FG-3DMOT does not rely on explicit and fixed assignments; instead, it flexibly assigns all detections to all objects simultaneously. The assignment issue is addressed implicitly and jointly with object state estimation by nonlinear least squares optimization. However, such methods ignore the consideration of the quality of detection and object prediction. For example, in cases of continuous missed detections, the predicted states can accumulate errors leading to inaccuracy, thereby affecting tracking performance. Additionally, there are some camera-LiDAR fusion-based multi-object tracking methods. In [25], researchers present a 3D multi-object tracking framework based on camera-LiDAR fusion with deep association, achieving seamless integration of 2 D and 3 D trajectories for more accurate and robust tracking, especially when tracked objects are far away. And in [26], researchers present CAMO-MOT, which introduces a combined appearance-motion optimization technique that significantly reduces tracking failures caused by occlusions and false detections by using camera and

在[22]中，研究人員提出 AB3DMOT，它是以 3D 多目標追蹤的基線為基礎，具有輕量級的架構和低計算需求，可實現即時操作。在[23]中，研究人員提出了一種新的 3D 多目標追蹤器，以預測置信度為導向進行資料關聯，可以更好地利用物體特徵進行點雲中的多目標追蹤。這兩種方法都是先進的純 LiDAR 即時 3D 多目標追蹤技術，它們都是利用 LiDAR 點雲資料所得到的偵測結果，並分別採用 Kalman filtering with Hungarian matching 與貪婪演算法來進行狀態估計與資料關聯。然而，如果資料關聯中的初始指派不正確，這種基於濾波的方法往往會陷入困境，很容易發生這種情況。在 [24] 中，研究人員提出一種基於因子圖的 3D 多目標追蹤 (FG-3DMOT) 方法，將偵測結果表示為高斯混合模型 (GMM)，並整合到因子圖架構中。FG-3DMOT 不依賴明確且固定的指派，而是靈活地將所有偵測結果同時指派給所有物件。分派問題是透過非線性最小二乘最佳化與物件狀態估計共同隱含處理。然而，這些方法忽略了偵測與物件預測品質的考量。舉例來說，在連續遺漏偵測的情況下，預測的狀態會累積誤差，導致不準確，進而影響追蹤效能。此外，還有一些基於攝影機-LiDAR 融合的多目標追蹤方法。在 [25] 中，研究人員提出了一個 3D 多目標追蹤框架，該框架基於攝影機-LiDAR 與深度聯想的融合，實現了 2D 與 3D 軌跡的無縫整合，從而實現更精確、更穩健的追蹤，尤其是當追蹤的物體距離較遠時。而在 [26] 中，研究人員提出了 CAMO-MOT，介紹了一種結合外觀與動作的最佳化技術，藉由使用攝影機與 LiDAR，可大幅減少因遮蔽與錯誤偵測所造成的追蹤失敗。

C. Coupled SLAM and Object Tracking

C. 耦合 SLAM 與物件追蹤

Currently, there have been many research efforts in coupled SLAM and object tracking. Specifically, researchers perform separate SLAM and object tracking systems simultaneously and estimate the ego-vehicle state and object state with a loosely-coupled approach [27]. In [28], visual odometry and persistent tracking are performed simultaneously to detect humans in dynamic environments. Additionally, in [15], researchers use neural networks to estimate object motion or segment potential dynamic objects from LiDAR point clouds, and SLAM is performed after filtering out dynamic object points. Another category integrates vehicle state estimation and object tracking into a unified optimization problem for tightly coupled SLAM with object tracking, which mainly revolve around visual-based and range sensor-based methods. In [29], researchers propose a tightly-coupled approach for SLAMMOT based on stereo visual information. In [5], Dynamic SLAM is proposed, which performs object detection

目前，在耦合 SLAM 與物體追蹤方面已有許多研究工作。具體來說，研究人員同時執行獨立的 SLAM 與物體追蹤系統，並以鬆散耦合的方式估算自我車輛狀態與物體狀態 [27]。在 [28] 中，同時執行視覺里程計與持續追蹤來偵測動態環境中的人類。此外，在 [15] 中，研究人員使用神經網路來估計物體的運動，或從 LiDAR 點雲中分割出潛在的動態物體，並在濾除動態物體點之後執行 SLAM。另一類是將車輛狀態估計與物體追蹤整合為一個統一的最佳化問題，以達到緊密耦合 SLAM 與物體追蹤的目的，主要圍繞著以視覺為基礎與以範圍感測器為基礎的方法。在 [29] 中，研究人員提出一種基於立體視覺資訊的 SLAMMOT 緊耦合方法。在 [5] 中，提出了動態 SLAM，可執行物體偵測

and tracking based on deep learning during the operation of SLAM in a dynamic environments. Furthermore, DynaSLAM II [6] demonstrates tightly-coupled multi-object tracking and visual SLAM through instance semantic segmentation and ORB features.

此外，DynaSLAM II [6] 透過實例語意分割與 ORB 特徵，展示緊密耦合的多目標追蹤與視覺 SLAM。此外，DynaSLAM II [6] 透過實例語意分割與 ORB 特徵，展示了緊密耦合的多目標追蹤與視覺 SLAM。

Regarding range sensor-based solutions, researchers in [4] propose a feasible tightly-coupled SLAMMOT system based on a laser scanner, deriving a dynamic Bayesian networks algorithm for coupling. In [30], researchers present an online SLAMMOT method using a grid-based approach, achieving pose estimation and object tracking in two-dimensional space. In [9], researchers present DL-SLOT, which uses sliding window-based graph optimization to simultaneously optimize the vehicle state and dynamic-static object states based on LiDAR. Based on DL-SLOT, the researchers present LIMOT [31], which is also a tightly-coupled multi-object tracking and LiDAR-inertial odometry system, and it presents a dynamic feature filtering method for the LiDAR odometry module. Although their introduced trajectory-based state prediction improves the tracking performance, this sliding window-based local optimization reduces its ability to perform global optimization for ego-vehicle and object states. In [10], researchers propose the IMM-SLAMMOT, which integrates interactive multiple model into object tracking for SLAMMOT to deal with the ambiguous motion patterns of objects. The data association in all above tracking methods follows fixed assignments, which tend to struggle if the initial assignment is incorrect. In [8], researchers propose LIO-SEGMOT for asynchronous estimation of

simultaneous ego-motion state and multiple object tracking, which integrates the factor graph based tracking that does not rely on explicit and fixed assignments in data association. However, these methods ignore considering the confidence during tracking, especially in scenarios with continuous missed detection due to occluded or distant objects, which can lead to performance degradation, affecting the accuracy and stability of SLAMMOT.

關於基於測距感測器的解決方案，研究人員在 [4] 中提出了一個可行的緊耦合 SLAMMOT 系統，該系統基於雷射掃描器，並衍生出一個動態貝葉斯網路演算法來進行耦合。在 [30] 中，研究人員提出一種線上 SLAMMOT 方法，使用基於網格的方法，在二維空間中實現姿態估計和物體追蹤。在 [9] 中，研究人員提出了 DL-SLOT，它使用基於滑動視窗的圖最佳化來同時最佳化車輛狀態和基於 LiDAR 的動靜物體狀態。在 DL-SLOT 的基礎上，研究人員提出了 LIMOT [31]，這也是一個緊密耦合的多物體追蹤與 LiDAR 慣性里程測量系統，並針對 LiDAR 里程測量模組提出了動態特徵篩選方法。雖然他們所引進的基於軌跡的狀態預測改善了追蹤性能，但這種基於滑動視窗的局部最佳化降低了其對自我車輛和物體狀態進行全局最佳化的能力。在 [10] 中，研究人員提出 IMM-SLAMMOT，將交互式多重模型整合至 SLAMMOT 的物件追蹤中，以處理物件模糊的運動模式。上述所有追蹤方法的資料關聯都是依據固定的分派，如果初始分派不正確，往往會很吃力。在 [8] 中，研究人員提出 LIO-SEGMOT 用於同步自我運動狀態的異步估計與多重物體追蹤，它整合了以因子圖為基礎的追蹤，在資料關聯時不依賴明確且固定的分配。然而，這些方法都忽略了追蹤過程中的信賴度考量，尤其是在因物體遮蔽或距離較遠而持續漏測的情境下，可能會導致性能下降，影響 SLAMMOT 的精確度與穩定性。

III. THE PROPOSED APPROACH

III.建議的方法

A. Overview of the Proposed Framework

A.建議架構概述

The proposed Conf SLAMMOT solution extends from the foundation of LiDAR SLAM based on graph optimization. The architecture of the Conf SLAMMOT solution is shown in Fig. 2, which mainly consists of an alternative LiDAR odometry module, an alternative moving object detection module, and a joint graph optimization backend, which is a tightly-coupled LiDAR SLAM and multi-object tracking method in dynamic environments for simultaneously estimating the states of the ego-vehicle and moving objects. The 3D point cloud information serves as the input for the LiDAR odometry module and the moving object detection network, respectively. The LiDAR SLAM method (the module used here is from LeGOLOAM [1]), as shown in Sec. III-B, consists of a replaceable LiDAR odometry module and partial graph optimization for the joint optimization backend of the overall system. For object detection, the advanced deep learning-based method, PV-RCNN [32], is used, as shown in Sec. III-C. As for the object tracking part of the system, inspired by FG-3DMOT

所提出的 Conf SLAMMOT 解決方案是在圖形最佳化的 LiDAR SLAM 基礎上延伸出來的。Conf SLAMMOT 解決方案的架構如圖 2 所示，主要由另一個 LiDAR 測距模組、另一個移動物體偵測模組及一個聯合圖最佳化後端所組成，是一種在動態環境下緊密耦合的 LiDAR SLAM 及多物體追蹤方法，可同時估計自我車輛及移動物體的狀態。三維點雲資訊分別作為 LiDAR 測距模組與移動物體偵測網路的輸入。如 Sec. III-B 所示，LiDAR SLAM 方法（此處使用的模組來自 LeGOLOAM [1]）由可替換的 LiDAR 測距模組和局部圖最佳化組成，為整體系統的聯合最佳化後端。至於物體偵測部分，則採用先進的深度學習方法 PV-RCNN [32]，如第三章 C 節所示。至於系統中的物體追蹤部分，則受 FG-3DMOT

[24], we use a max-mixture model for implicitly associating object prediction states with detection results and integrate it with a prediction confidence model for confidence-guided data association. To demonstrate the improvements we have made in data association, we display this implicit data association as a module in the figure. It should be noted that this implicit data association is managed as part of the factor graph optimization and is to be jointly optimized with the overall system, as shown in Sec. III-D and E. Finally, the joint factor graph optimization backend outputs the state estimation of the ego-vehicle and moving objects

simultaneously, as shown in Sec. III-F.

[24]，我們使用 max-mixture 模型將物件預測狀態與偵測結果隱含地聯繫起來，並將其與預測置信度模型整合，以進行置信度引導的資料聯繫。為了展示我們在資料關聯方面的改進，我們將此隱含資料關聯顯示為圖中的一個模組。值得注意的是，這個隱含的資料關聯是作為因素圖最佳化的一部分來管理，並且要與整體系統共同最佳化，如第 III-D 和 E 節所示。最後，共同因素圖最佳化後端會同時輸出自我車輛和移動物體的狀態估計，如第 III-F 節所示。

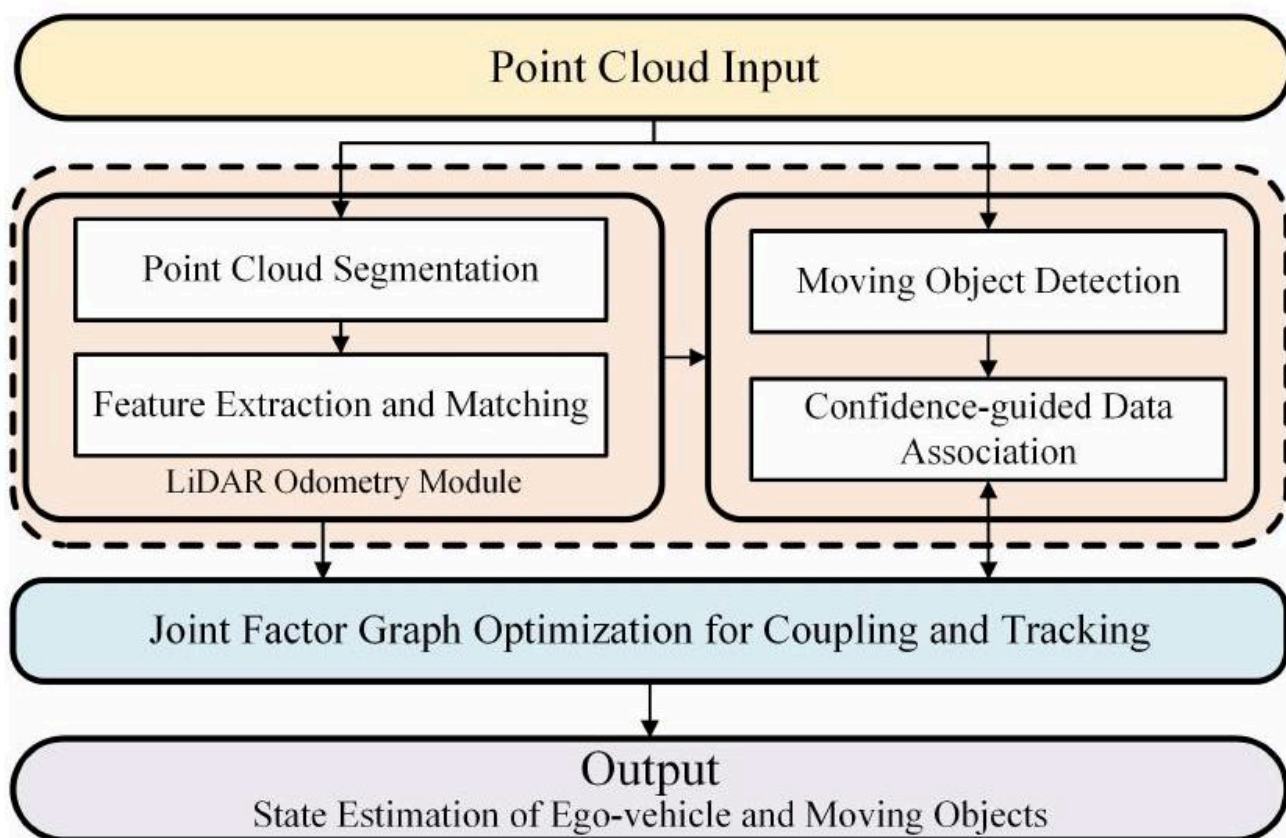


Fig. 2. Overall architecture of the presented Conf SLAMMOT solution.

圖 2.Conf SLAMMOT 解決方案的整體架構。

B. LiDAR Odometry Module

B.LiDAR 航道測量模組

In presented Conf SLAMMOT solutions, the SLAM can be any representative LiDAR SLAM framework. Due to the several sub-modules of the proposed system, the lightweight of each module should be considered under conditions of limited computing resources. Also, the IMU is not a mandatory sensor in many cases. Therefore, LeGO-LOAM [1] is applied for LiDAR odometry module, which maintains high localization and mapping accuracy while consuming fewer computing resources compared to the LOAM [12].

在所提出的 Conf SLAMMOT 解決方案中，SLAM 可以是任何具代表性的 LiDAR SLAM 架構。由於建議系統有多個子模組，在有限的計算資源條件下，應考慮每個模組的輕量級。此外，IMU 在許多情況下並非必須使用的感測器。因此，LeGO-LOAM [1] 應用於 LiDAR 測距模組，相較於 LOAM [12]，LeGO-LOAM 可維持較高的定位與繪圖精準度，同時消耗較少的運算資源。

The LeGO-LOAM receives input from a 3D LiDAR and outputs a 6 Degrees of Freedom (DoF) ego-pose estimate and a 3D point cloud map. This framework is divided into five submodules: segmentation, feature extraction, LiDAR odometry, LiDAR mapping, and trajectory integration [1]. The LeGOLOAM project the raw point cloud into a distance image and differentiating between ground points and segmented points before feature extraction. It then matches feature points of the same category and incorporates the iSAM2 [33] optimizer in the backend. Finally, the ego-vehicle pose estimation obtained by this module will provide to the

multi-object tracking and joint graph optimization backend.

LeGO-LOAM 接收來自 3D LiDAR 的輸入，並輸出 6 自由度 (DoF) 的自我姿勢估計和 3D 點雲圖。此架構分為五個子模組：分割、特徵萃取、LiDAR 測距、LiDAR 繪圖與軌跡整合[1]。LeGOLOAM 將原始點雲投影為距離影像，並在特徵萃取前區分地面點與分割點。LeGOLOAM 將原始點雲投影成距離影像，並在特徵萃取前區分地面點與分割點，然後將相同類別的特徵點進行匹配，並在後端加入 iSAM2 [33] 最佳化器。最後，此模組所獲得的自我車輛姿勢估計將提供給多目標追蹤與聯合圖形最佳化後端。

C. Object Detection Module

C.物件偵測模組

For object detection in Conf SLAMMOT, it can be the any representative detection method. Herein, we use the PVRCNN, which is an advanced framework capable of achieving precise 3D object detection from point clouds with manageable memory consumption. PV-RCNN integrates the advantages of 3D voxel CNN with sparse convolution [34] and PointNetbased networks [35] to learn more discriminative point cloud

對於 Conf SLAMMOT 中的物件偵測，它可以是任何具有代表性的偵測方法。在此，我們使用 PVRCNN，它是一個先進的架構，能夠以可管理的記憶體消耗，從點雲達到精確的 3D 物體偵測。PV-RCNN 整合了三維體素 CNN 與稀疏卷積 [34] 及 PointNetbased 網路 [35] 的優點，以學習更具辨別力的點雲檢測方法。

features. The method utilizes a 3D voxel sparse CNN as the backbone network and employs a voxel set abstraction module to convert 3D scenes into keypoints, followed by a keypoint-to-grid Region of Interest (RoI) feature abstraction for refinement, used for proposal confidence prediction and location refinement. The two-step strategy of this framework effectively combines voxel-based and point-based learning, demonstrating accuracy and efficiency in 3D object detection using point cloud data without additional sensor inputs. PVRCNN's innovative approach and its performance on the KITTI benchmark have established it as a popular solution in the field of autonomous driving object detection [11].

特徵。該方法利用 3D 象素稀疏 CNN 作為骨幹網路，並採用象素集抽象模組將 3D 場景轉換為關鍵點，接著再以關鍵點到網格的區域興趣 (RoI) 特徵抽象進行精細化，用於提案置信度預測和位置精細化。此架構的兩步策略有效結合了以象素為基礎的學習與以點為基礎的學習，在使用點雲資料進行 3D 物體偵測時，無須額外的感測器輸入，即能展現精確度與效率。PVRCNN 的創新方法及其在 KITTI 基準上的表現，使其成為自主駕駛物體偵測領域的熱門解決方案[11]。

D. Object State Prediction

D.物件狀態預測

For object state prediction, considering prediction accuracy and efficiency, most current 3D object detection algorithms typically employ motion models to predict future object states, with constant velocity (CV) [22], [36], [37], constant acceleration (CA) [23], and constant turn rate and velocity (CTRV) models [38] being widely utilized. The CV model assumes that the object moves at a constant velocity, disregarding the effects of acceleration. This model is particularly effective when object motion is smooth, without sharp turns or changes in acceleration. The CA model assumes the object moves with constant acceleration, making it suitable for scenarios where the object's acceleration is relatively stable. The CTRV model assumes the object maintains a constant turn rate and velocity during turning, which is applicable to cases involving curved motion. Additionally, some learning-based tracking methods explore predicting states by learning object position transformations from historical trajectory data through neural networks [39]-[41]; however, they are typically several times slower than the aforementioned motion models and often face challenges in practical applications. Therefore, taking into account both prediction accuracy and efficiency, as well as addressing the missed detection issue we discuss, where objects are more likely to be lost during turning scenarios, we adopt the CTRV model to predict the future states of objects.

對於物體狀態預測，考慮到預測準確度與效率，目前大多數的 3D 物體偵測演算法通常都會採用運動模型來預測未來的物體狀態，其中恆定速度 (CV) [22]、[36]、[37]、恆定加速度 (CA) [23] 以及恆定轉向速率與速度 (CTRV) 模型 [38] 被廣泛運用。CV 模型假設物件以固定速度移動，不考慮加速度的影響。當物件移動平順，沒有急轉彎或加速度變化時，此模型特別有效。CA 模型假設物件以固定加速度移動，因此適用於物件加速度相對穩定的情況。CTRV 模型假設物件在轉彎時保持恆定的轉彎速率和速度，適用於涉及曲線運動的情況。此外，有些以學習為基礎的追蹤方法是透過神經網路從歷史軌跡資料中學習物件位置轉換來探索預測狀態 [39]-[41]；然而，這些方法的速度通常比前述的運動模型慢上數倍，在實際應用上也經常面臨挑戰。因此，考量到預測準確度與效率，並針對我們所討論的遺漏偵測問題，也就是在轉彎情境中物體較容易遺失的問題，我們採用 CTRV 模型來預測物體未來的狀態。

After obtaining the object state predictions and detection results, most traditional methods directly associate them (explicitly or implicitly) to solve the assignment problem. However, the predicted states of objects are not always accurate; for instance, after missing detections in several consecutive frames, they can accumulate significant errors, thus enlarging the possible range of detected objects when resuming tracking. Meanwhile, the quality of detection results also needs to be considered. Therefore, we have incorporated a prediction confidence model [23] into our present system to make data association more sensitive to the quality of each prediction.

在獲得物件狀態預測與偵測結果後，大多數傳統方法會直接將兩者聯繫起來（顯式或隱式）以解決分配問題。然而，預測的物體狀態並不總是精確的，例如，在連續幾個幀中遺漏偵測之後，就會累積相當大的誤差，因而在重新開始追蹤時會擴大可能偵測到的物體範圍。同時，偵測結果的品質也需要考慮。因此，我們在目前的系統中加入了預測置信度模型 [23]，讓資料關聯對每個預測的品質更加敏感。

$$\hat{c}_{pre,k}^i = c_{pre,k}^i - \alpha c_{pre,k-1}^i$$

Here, $c_{pre} \in (0, 1]$ denotes the prediction confidence and $\alpha \in [0, 1]$ denotes a parameter that adjusts the overall impact of prediction confidence on data association and is determined by dataset. Applying this model can address the issue of continuous multi-frame missed detections, where when a miss occurs, the prediction confidence decreases, and the implicit search range for data association is subsequently enlarged.

這裡， $c_{pre} \in (0, 1]$ 表示預測置信度， $\alpha \in [0, 1]$ 表示調整預測置信度對資料關聯的整體影響的參數，該參數由資料集決定。應用此模型可以解決連續多畫面遺漏偵測的問題，當遺漏發生時，預測置信度會降低，資料關聯的隱含搜尋範圍也會隨之擴大。

This flexible adjustment of the implicit search range for data association can better recover re-tracking after continuous missed detection, preventing ID switches.

這種對資料關聯的隱含搜尋範圍的靈活調整，可以更好地恢復連續遺漏檢測後的重新追蹤，防止 ID 切換。

E. Confidence-guided Implicit Data Association

E. 信心引導的隱含資料關聯

The designed implicit confidence-based data association solves the assignment problem between detection results and predicted states with their corresponding confidence. This method employs the GMM to simultaneously represent all detection results, and we utilize the max-mixture model [24], [42] to represent the full GMM of each object that is integrated with the prediction confidence model. Specifically, we apply the GMM as a probabilistic model of detection results that overcomes the limitations of simple single Gaussian models and breaks away from the least squares formulation inherent in the maximum likelihood problem, selecting one from all possible predicted states of $\bar{\mathbf{O}}_k$ such that the likelihood of observing the current data given the detection results $\mathbf{Z}_k$ is the highest. This confidence-guided implicit data association integrates the assignment problem into the factor optimization to jointly estimate the object state and solve the data association implicitly as part of the optimization, which allows for a flexible association between predicted states and detections that can change during the optimization process, unlike most online trackers, and goes beyond merely estimating the association by optimizing state positions across the entire sequence to account for inaccuracies in object prediction and detection. By applying the negative logarithm, the optimization

problem arising from the maximum likelihood formulation can be reformulated as follows [43]:

所設計的隱含式信賴資料關聯解決了偵測結果與預測狀態及其相對應的信賴度之間的分配問題。此方法運用 GMM 來同時表示所有偵測結果，我們利用最大混合模型 [24]、[42] 來表示與預測置信度模型整合的每個物件的完整 GMM。具體來說，我們應用 GMM 作為偵測結果的概率模型，它克服了簡單單一高斯模型的限制，並擺脫了最大似然問題中固有的最小二乘公式，從 $\bar{\mathbf{O}}_k$ 的所有可能預測狀態中選擇一個，使得在偵測結果 $\mathbf{Z}_k$ 的情況下，觀察到目前資料的可能性最高。這種置信引導的隱式資料關聯將指派問題整合到因素最佳化中，以共同估計物體狀態，並將資料關聯隱式地作為最佳化的一部分來求解，這使得預測狀態與偵測之間的關聯具有彈性，可以在最佳化過程中改變，這與大多數線上追蹤器不同，而且不僅僅是估計關聯，還會在整個序列中最佳化狀態位置，以考量物體預測與偵測的不精確性。運用負對數，最大似然公式所產生的最佳化問題可以重新表達如下[43]：

$$\hat{\mathbf{O}} = \arg \min_{\mathbf{O}} \sum_i -\ln \left(\mathbf{P} \left(\mathbf{z}_{i,k} \mid \bar{\mathbf{O}}_{i,k} \right) \right)$$

Where the optimal estimated $\hat{\mathbf{O}}_k$ denotes the maximumlikelihood estimator of $\bar{\mathbf{O}}_k$; $\mathbf{P} \left(\mathbf{z}_{i,k} \mid \bar{\mathbf{O}}_{i,k} \right)$ denotes the conditional probability that associates the i -th predicted object state $\bar{\mathbf{O}}_{i,k}$ with the corresponding detection result $\mathbf{z}_{i,k}$ that belongs to it. For a sum of Gaussians with n components (c), the conditional probability $\mathbf{P}$ is defined as:

其中，最佳估計值 $\hat{\mathbf{O}}_k$ 表示 $\bar{\mathbf{O}}_k; \mathbf{P} \left(\mathbf{z}_{i,k} \mid \bar{\mathbf{O}}_{i,k} \right)$ 的最大似然估計值， $\bar{\mathbf{O}}_k; \mathbf{P} \left(\mathbf{z}_{i,k} \mid \bar{\mathbf{O}}_{i,k} \right)$ 表示將 i -th 預測物件狀態 $\bar{\mathbf{O}}_{i,k}$ 與屬於它的相應偵測結果 $\mathbf{z}_{i,k}$ 相關聯的條件概率。對於具有 n 分量 (c) 的高斯總和，條件概率 $\mathbf{P}$ 定義為：

$$\mathbf{P} \left(\mathbf{z}_{i,k} \mid \bar{\mathbf{O}}_{i,k} \right) \propto \sum_{j=1}^n c_j \cdot \exp \left(-\frac{1}{2} \left\| \Sigma_j^{-\frac{1}{2}} \left(\bar{\mathbf{O}}_{i,k} - \mathbf{T}_{E,k}^{M_E} \cdot \mathbf{z}_{j,k} \right) \right\|^2 \right)$$

Herein, m denotes the sum of detection measurements; j denotes the j th detection. $\mathbf{T}_{E,k}^{M_E}$ denotes the transformation from ego-vehicle coordinate system to map coordinate system M_E . And $c_j = \omega_j \cdot \det \left(\Sigma_j^{-\frac{1}{2}} \right)$, where ω_j and Σ_j denotes the weight and uncertainty of j th detection. Herein, the Σ_j considers the confidence score of detection results, denoting $\Sigma_j = \Gamma \cdot \left(1 - c_{det,k}^j \right) \cdot \beta$, instead of using the overall uncertainty of the detector Γ determined by sensor's error characteristic. $c_{det,k}$ denotes the confidence score of the corresponding detection box, and β denotes a scaling parameter that can vary depending on the detector and the dataset. And for the conditional probability $\mathbf{P}$, the logarithm cannot be pushed inside due to the summation, so we need to calculate the log-likelihood differently. There are two suggested solutions: an approximate max-mixture method [42] and an exact sum-mixture method [44]. And the max-mixture approach simplifies calculations compared to the sum-of-Gaussians method while maintaining similar effectiveness. Therefore, the summation of a GMM can

在此， m 表示偵測量測的總和； j 表示第 j 次偵測。 $\mathbf{T}_{E,k}^{M_E}$ 表示從自我車輛坐標系到地圖坐標系 M_E 的轉換。而 $c_j = \omega_j \cdot \det \left(\Sigma_j^{-\frac{1}{2}} \right)$ ，其中 ω_j 和 Σ_j 表示第 j 次偵測的權重和不確定性。在此， Σ_j 考慮的是偵測結果的置信分數，表示為 $\Sigma_j = \Gamma \cdot \left(1 - c_{det,k}^j \right) \cdot \beta$ ，而不是使用由感測器的誤差特性所決定的偵測器整體不確定性 Γ 。 $c_{det,k}$ 表示相對應檢測方塊的置信度分數， β 表示可根據檢測器和資料集而變化的比例參數。而對於條件概率 $\mathbf{P}$ ，由於求和的關係，對數無法推到裡面，因此我們需要以不同的方式計算 log-likelihood。有兩種建議的解決方法：近似最大混合法 [42] 和精確總和混合法 [44]。與高斯總和法相比，最大混合法簡化了計算，同時保持了相似的效果。因此，GMM 的求和可以

be replaced by the maximum operator as following equation (4), and we push inside the logarithm for maximum becoming a minimum as shown in equation (5):

由最大值算子取代，如下式 (4)，我們在對數內推最大值成為最小值，如式 (5) 所示：

$$-\ln(\mathbf{P}) = -\ln \max_j \left(c_j \cdot \exp \left(-\frac{1}{2} \left\| \Sigma_j^{-\frac{1}{2}} \left(\overline{\mathbf{O}}_{i,k} - \mathbf{T}_{E,k}^{M_E} \cdot \mathbf{Z}_{j,k} \right) \right\|^2 \right) \right)$$

$$-\ln(\mathbf{P}) = \min_j \left(-\ln c_j + \frac{1}{2} \left\| \Sigma_j^{-\frac{1}{2}} \left(\overline{\mathbf{O}}_{i,k} - \mathbf{T}_{E,k}^{M_E} \cdot \mathbf{Z}_{j,k} \right) \right\|^2 \right)$$

To maintain a uniform surface in the optimization formula, we perform dimensional separation between the logarithmic normalization term and the quadratic error term. This not only aids in achieving better convergence but also allows us to consider the quadratic error term individually since the lognormalization term is a constant. The least squares problem in this optimization is expressed as follows:

為了在最佳化公式中維持均勻的表面，我們在對數歸一化項和二次方誤差項之間進行了分離。這不僅有助於達到更好的收斂效果，而且由於對數歸一化項是常數，因此我們可以單獨考慮二次誤差項目。此最佳化中的最小二乘問題表達如下：

$$\|\mathbf{e}_{i,k}\|^2 = \min_j \left\| \frac{\sqrt{-2 \cdot \ln \frac{c_j}{c_{\max}}}}{\Sigma_j^{-\frac{1}{2}} \left(\overline{\mathbf{O}}_{i,k} - \mathbf{T}_{E,k}^{M_E} \cdot \mathbf{Z}_{j,k} \right)} \right\|^2$$

The term $-\ln \frac{c_j}{c_{\max}}$ represents the independent dimension of the vectorized error function, where $c_{\max}$ is a normalization constant to ensure that the expression under the square root is positive. $\mathbf{e}$ denotes the weighted error between the i th object state and j th detection measurement. Then, considering the prediction confidence model in equation (1), the quadratic error term can be rewritten as follows:

項 $-\ln \frac{c_j}{c_{\max}}$ 代表向量化誤差函數的獨立維度，其中 $c_{\max}$ 為歸一常數，以確保平方根下的表達式為正值； $\mathbf{e}$ 表示 i 第 3 個物體狀態與 j 第 3 個偵測測量之間的加權誤差。然後，考慮到公式 (1) 中的預測置信度模型，二次誤差項可重寫如下：

$$\hat{c}_{pre,k}^i \cdot \min \left\| \Sigma_i^{-\frac{1}{2}} \left(\overline{\mathbf{O}}_{i,k} - \mathbf{T}_{E,k}^{M_E} \cdot \mathbf{Z}_{i,k} \right) \right\|^2 < \sigma$$

Herein, $\sigma > 0$ denotes an association threshold to determine whether the detection belongs to any existing object with prediction confidence. Specifically, for each predicted state, a maximum likelihood model is employed to correlate all detection results, and the minimum value of the aforementioned formula is selected. If this minimum value is less than the threshold, the detection is considered associated. Compared to approaches that do not take into account the prediction confidence, this method effectively enlarges the implicit range of association, which is highly effective for resuming tracking after consecutive missed detections, as shown in Fig. 3(b).

其中， $\sigma > 0$ 表示關聯閾值，用以判斷偵測結果是否屬於任何具有預測可信度的現有物件。具體而言，對於每個預測狀態，採用最大似然模型來關聯所有偵測結果，並選擇前述公式的最小值。若此最小值小於臨界值，則視為相關的偵測結果。如圖 3(b) 所示，與不考慮預測置信度的方法相比，此方法有效地擴大了關聯的隱含範圍，對於在連續遺漏偵測後恢復追蹤非常有效。

The aforementioned implicit data association is integrated into graph optimization, after which the tracking state for matched detection states and predicted states are updated by using the optimization, in order to obtain the updated object states. Concurrently, the prediction confidence, based on the confidence of the corresponding detection results $c_{det,k}$, will also be updated as follows:

將前述隱含資料關聯整合至圖表最佳化，再利用最佳化更新匹配偵測狀態與預測狀態的追蹤狀態，以獲得更新後的物件狀態。同時，根據對應的偵測結果 $c_{det,k}$ 的信賴度，也會更新預測信賴度，如下所示：

$$c_{pre,k}^m = \begin{cases} 1, & c_{pre,k-1}^m = 0, c_{pre,k}^m \neq 0 \\ \hat{c}_{pre,k}^m + \alpha c_{det,k}^m, & \text{otherwise} \end{cases}$$

where m denotes the matched paris; $c_{pre,k-1}^m$ denotes the confidence of missed detection at the moment $k - 1$.

其中 m 表示匹配的 Paris ; $c_{pre,k-1}^m$ 表示在 $k - 1$ 時遺漏偵測的置信度。

For unmatched detection and prediction results, we employ the following tracking management strategy. Detection results with very low confidence are directly discarded, while those with higher confidence are initialized to establish new tracks. For unmatched predicted object states, there are two scenarios. The first scenario is when the object naturally disappears from the field of view. The second scenario occurs in challenging situations where the object is temporarily occluded by other objects or is too far from the sensor, leading to missed detections. We set a threshold N for the number of prediction frames to distinguish between these two cases. If a predicted

對於不匹配的偵測與預測結果，我們採用以下的追蹤管理策略。置信度極低的偵測結果會被直接捨棄，而置信度較高的偵測結果則會被初始化以建立新的軌跡。對於不匹配的預測物件狀態，有兩種情況。第一種情況是物體從視野中自然消失。第二種情況發生在具有挑戰性的情況下，即物體暫時被其他物體遮擋或距離感測器太遠，導致偵測遺漏。我們設定了預測幀數的臨界值 N ，以區分這兩種情況。如果預測的

state cannot be updated by detection states for more than N frames (set to 12 here), it is considered to have naturally vanished. Such predicted states are removed and no longer tracked. Otherwise, the predicted states are retained, as the object may have been temporarily missed and could reappear in future frames, as illustrated in Fig. 3 bb), where the state performs prediction from time $k - t$ to $k - 1$ and re-associates data guided by confidence at time k to resume tracking.

在超過 N 幀（此處設為 12 幀）的時間內，狀態無法被偵測狀態更新，則視為已自然消失。這種預測狀態會被移除，不再被追蹤。否則，預測的狀態會被保留，因為物體可能暫時被遺漏，並可能在未來的幀中再次出現，如圖 3 bb) 所示，其中狀態會從時間 $k - t$ 到 $k - 1$ 執行預測，並在時間 k 重新關聯由置信度引導的資料，以恢復追蹤。

F. Joint Factor Graph Optimization

F.聯合因子圖最佳化

The unified graph optimization backend couples SLAM with the multi-object tracking to jointly optimize the state estimation of the ego-vehicle and moving objects. It extends the factor graph optimization of the LiDAR SLAM module and is integrated with factor graph-based tracking. As depicted in Fig. 3 (a), all variable nodes represent the estimated 3D states. The node corresponding to the ego-vehicle pose $\mathbf{X}_{E,k^*}$ at the timestamp of selected key frames k^* is obtained through LiDAR odometry. The edge connecting two nodes represents the ego-vehicle motion constraint factor $\mathbf{e}_{odo}$:

統一圖形最佳化後端將 SLAM 與多物體追蹤結合在一起，以共同最佳化自我車輛與移動物體的狀態估計。它擴充了 LiDAR SLAM 模組的因果圖最佳化，並整合了基於因果圖的追蹤。如圖 3 (a) 所示，所有變數節點代表估計的 3D 狀態。在選取的關鍵幀 k^* 的時間戳中，對應於自我車輛姿勢 $\mathbf{X}_{E,k^*}$ 的節點是透過 LiDAR 測距獲得的。連接兩個節點的邊緣代表自我車輛運動限制因子 $\mathbf{e}_{odo}$ ：

$$\mathbf{e}_{odo}^{k^*-1,k^*}(\mathbf{X}_{k^*-1}, \mathbf{X}_{k^*}) = \mathbf{T}_{k^*-1}^{k^*} \mathbf{X}_{k^*-1} - \mathbf{X}_{k^*}$$

The object perception factor connects the ego-vehicle pose node and the i -th moving object pose node $\mathbf{O}_{i,k}$ that deals with the implicit data association between prediction objects and detection measurements by the confidence-guided implicit data association introduced in Sec. III-D, as shown in Fig. 3 (b). For computational efficiency in LiDAR odometry, keyframe selection is used for backend state estimation, meaning that the vehicle state is updated only in keyframes, and nonkeyframe LiDAR data, including moving object detections, is ignored. Conventional methods typically estimate vehicle and object states synchronously, but factor graph optimizationbased solutions do not have this limitation. Inspired by [8], we use asynchronous state estimation to utilize object information from non-keyframes. This involves transforming object detection results from non-keyframes to virtual moving object detection measurements in the latest keyframes, as shown in Fig. 3(c). This transformation uses the relative vehicle state estimates at timestamps k and k^* through scan matching.

Such an object perception factor is denoted as:

物體感知因子連接自我車輛姿勢節點和 i -th 移動物體姿勢節點 $\mathbf{O}_{i,k}$ ，該節點透過 Sec. III-D 中介绍的置信引導隱含資料關聯來處理預測物體和偵測測量之間的隱含資料關聯，如圖 3 (b) 所示。為了提高 LiDAR 測距的計算效率，後端狀態估計採用關鍵影格選擇，也就是僅在關鍵影格中更新車輛狀態，而忽略非關鍵影格的 LiDAR 資料，包括移動中的物體偵測。傳統的方法通常會同步估計車輛與物件的狀態，但以因數圖最佳化為基礎的解決方案則沒有這個限制。受到 [8] 的啟發，我們使用異步狀態估計來利用非關鍵影像的物件資訊。這包括將非關鍵影格的物件偵測結果轉換為最新關鍵影格中的虛擬移動物件偵測量測，如圖 3(c) 所示。此轉換使用時間戳 k 和 k^* 時透過掃描比較的相對車輛狀態估計。這樣的物件感知係數表示為：

$$\mathbf{e}_{op}^{k^*,k}(\mathbf{X}_{E,k^*}, \mathbf{O}_{i,k} | \mathbf{z}_{i,k}) = \mathbf{O}_{i,k}^{M_E} - \mathbf{X}_{E,k^*}^{M_E} \mathbf{z}_{i,k}$$

The edge between two object pose nodes is denoted as the object motion constraint factor $\mathbf{e}_{mov,i}^{k-1,k}$ shown in equation (11), where $f(\cdot)$ denotes the CTRV movement function of object states driven by its velocity $\mathbf{V}_{i,k}$ in a time step ΔT , where the $\mathbf{V}_{i,k} = [\mathbf{v}_{i,k}, \omega_{i,k}]^T$ denotes the linear velocity and angular velocity respectively. The edge constraints between two poses can be represented by velocity factors $\mathbf{e}_{v,i}^{k-1,k}$ in equation (12), which assume a reasonable approximation that the object moves with a constant velocity over a short time.

兩個物件姿態節點之間的邊用等式 (11) 所示的物件運動約束係數 $\mathbf{e}_{mov,i}^{k-1,k}$ 表示，其中 $f(\cdot)$ 表示在時間步驟 ΔT 中，物件狀態由其速度 $\mathbf{V}_{i,k}$ 驅動的 CTRV 運動函數，其中 $\mathbf{V}_{i,k} = [\mathbf{v}_{i,k}, \omega_{i,k}]^T$ 分別表示線性速度和角速度。兩個姿態之間的邊緣約束可以用公式 (12) 中的速度係數 $\mathbf{e}_{v,i}^{k-1,k}$ 來表示，其中假設一個合理的近似值，即物件在短時間內以固定速度移動。

$$\begin{aligned} \mathbf{e}_{mov,i}^{k-1,k}(\mathbf{O}_{i,k-1}, \mathbf{O}_{i,k}, \mathbf{V}_{i,k-1}) &= f(\mathbf{O}_{i,k-1}, \mathbf{V}_{i,k-1}, \Delta T) - \mathbf{O}_{i,k} \\ \mathbf{e}_{v,i}^{k-1,k}(\mathbf{V}_{i,k-1}, \mathbf{V}_{i,k}) &= \mathbf{V}_{i,k-1} - \mathbf{V}_{i,k} \end{aligned}$$

Finally, the joint factor graph optimization is defined as

最後，聯合因素圖最佳化定義為

minimizing the sum of nonlinear least-square errors:

最小化非線性最小平方誤差之和：

$$\begin{aligned} X^* = \arg \min_{\chi} & \sum_{k^*-1, k^*} \left\| \mathbf{e}_{odo}^{k^*-1, k^*} \right\|_{\Sigma_{odo}}^2 + \left\| \mathbf{e}_p(\chi_0) \right\|^2 \\ & + \sum_{\substack{i, k-1 \\ k^*, k}} \left(\left\| \mathbf{e}_{op}^{k^*, k} \right\|_{\Sigma_{op}}^2 + \left\| \mathbf{e}_{mov,i}^{k-1, k} \right\|_{\Sigma_{mov}}^2 + \left\| \mathbf{e}_{v,i}^{k-1, k} \right\|_{\Sigma_v}^2 \right) \end{aligned}$$

Where χ denotes the set of all variables, and $\left\| \mathbf{e}_p(\chi_0) \right\|^2$ denotes the prior information error term. Σ_{odo} , Σ_{op} , Σ_{mov} , Σ_v , and Σ_p denote the covariance or standard deviation matrix of each variable factor, respectively.

其中 χ 表示所有變數的集合， $\left\| \mathbf{e}_p(\chi_0) \right\|^2$ 表示先驗資訊誤差項。 Σ_{odo} , Σ_{op} , Σ_{mov} 、 Σ_v 和 Σ_p 分別表示各變數因子的協方差或標準差矩陣。

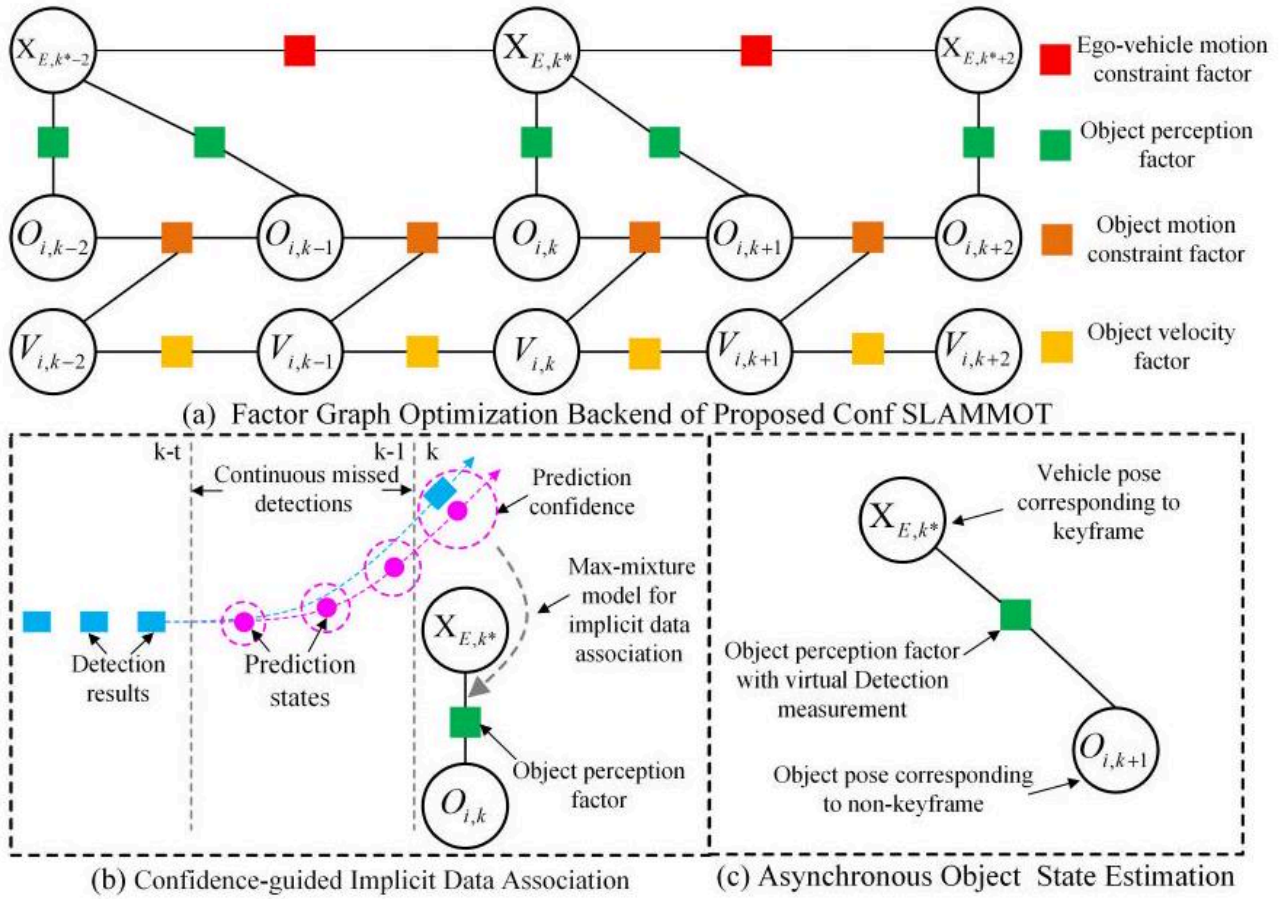


Fig. 3. The factor graph model in proposed Conf SLAMMOT solution. (a) Joint factor graph optimization backend for coupling and tracking. (b) is an explanation subfigure of confidence-guided implicit data association. (c) is an explanation subfigure of asynchronous object state estimation.

圖 3. Conf SLAMMOT 解決方案中的因子圖模型。(a) 用於耦合和追蹤的聯合因子圖最佳化後端。(b) 是信心導向隱含資料關聯的解釋子圖。(c) 是異步物件狀態估計的解釋子圖。

IV. EXPERIMENTAL EVALUATION

IV. 實驗評估

A. Experiment Conditions

A. 實驗條件

We provide a comprehensive overview of the experiments conducted to evaluate the performance of the proposed Conf SLAMMOT and the baseline methods in various aspects. We conducted experiments on several sequences of the KITTI Tracking dataset [11]. This dataset was primarily collected on urban roads and highways, and it includes raw point clouds, IMU/GPS data, and ground truth labels that can be used for tracking evaluation. We use 18 sequences of dataset with ground truth label data as evaluation sequences, excluding sequences 16, 17, and 20, because they are either highway sequences that cause SLAM to fail or sequences that primarily focus on pedestrian objects. To highlight the coupling advantages of the Conf SLAMMOT over standalone LiDAR SLAM or multi-object tracking systems, we implement comparative experiments by several methods to evaluate our proposed solution for state estimation of ego-vehicle and objects. And we also evaluate the overall performance of Conf SLAMMOT with other baseline methods in different sequences. For the parameters and thresholds used in confidence-guided implicit data association, we set α to 0.03, β to 80 and σ to 6.5 for the applied method and dataset. Additionally, we evaluate the time

consumption of each module in the proposed solution.

我們全面概述了為評估所提出的 Conf SLAMMOT 與基線方法在各方面的效能而進行的實驗。我們在 KITTI Tracking 資料集 [11] 的幾個序列上進行實驗。這個資料集主要是在城市道路和高速公路上收集的，它包含原始點雲、IMU/GPS 資料和可用做追蹤評估的地面真實標籤。我們使用資料集中的 18 個序列與地面真實標籤資料作為評估序列，但不包括序列 16、17 與 20，因為它們不是會導致 SLAM 失敗的高速公路序列，就是主要針對行人物件的序列。為了突顯 Conf SLAMMOT 相較於獨立 LiDAR SLAM 或多目標追蹤系統的耦合優勢，我們使用多種方法進行比較實驗，以評估我們所提出的自我車輛與物體狀態估計解決方案。此外，我們也評估了 Conf SLAMMOT 與其他基線方法在不同序列中的整體效能。對於置信引導的隱含資料關聯所使用的參數和臨界值，我們針對應用的方法和資料集，設定 α 到 0.03, β 為 80, σ 為 6.5。此外，我們還評估了建議解決方案中每個模組的耗時。

B. Ego-vehicle Pose Estimation Evaluation

B.自我車輛姿勢評估

In the ego-pose estimation evaluation experiments, we compare the pose estimation among three methods. Considering

在自我姿勢估計評估實驗中，我們比較了三種方法的姿勢估計。考慮到

the potential different outcomes of whether to filter out dynamic object points during LiDAR odometry for scan matching, LOAM [12], LeGO-LOAM [1], and Conf SLAMMOT represent the methods that perform dynamic point filtering, while LOAM*, LeGO-LOAM*, and Conf SLAMMOT* indicate the methods that do not perform dynamic point filtering. We use the mean error (MEAN) and root mean square error (RMSE) as evaluation metrics. The experimental results in various sequences of the KITTI Tracking dataset are shown in Table I. It can be observed from the table that in a few individual sequences, the performance after filtering out dynamic points is inferior to that without such filtering. This is because points belonging to potential dynamic objects (such as temporarily parked vehicles, etc.) are removed, leading to feature sparsity and thus reducing the pose estimation accuracy. For most scenarios with many moving objects and few potential dynamic objects, filtering out dynamic points to ensure the reliability of feature points can enhance pose estimation accuracy. Moreover, the pose estimation accuracy of the Conf SLAMMOT is generally higher than that of LOAM and LeGO-LOAM, indicating that in dynamic environments, tightly coupling SLAM and MOT in the graph optimization backend can benefit each other, as for SLAM, moving objects are no longer treated as interfering information but as auxiliary information to enhance the ego-pose estimation. Additionally, Fig. 4 displays the ego-trajectory error maps for several sequences, providing a more intuitive comparison.

LOAM [12]、LeGO-LOAM [1] 與 Conf SLAMMOT 代表執行動態點過濾的方法，而 LOAM*、LeGO-LOAM* 與 Conf SLAMMOT* 則表示不執行動態點過濾的方法。我們使用平均誤差 (MEAN) 和均方根誤差 (RMSE) 作為評估指標。在 KITTI Tracking 資料集的各種序列中的實驗結果如表 I 所示。從表中可以觀察到，在少數的個別序列中，濾除動態點之後的效能比沒有濾除動態點的效能還差。這是因為屬於潛在動態物件（如暫時停放的車輛等）的點被移除，導致特徵稀疏，進而降低姿勢估計的準確度。對於大多數移動物件多、潛在動態物件少的場景，濾除動態點以確保特徵點的可靠性，可以提高姿勢估計的準確度。此外，Conf SLAMMOT 的姿勢估計準確度普遍高於 LOAM 與 LeGO-LOAM，顯示在動態環境下，將 SLAM 與 MOT 在圖最佳化後端緊密耦合，可以互惠互利，因為對 SLAM 而言，移動物體不再被視為干擾資訊，而是增強自我姿勢估計的輔助資訊。此外，圖 4 顯示幾個序列的自我軌跡誤差圖，提供更直覺的比較。

TABLE I 表一
MEAN/RMSE OF EGO-POSE ESTIMATION ON KITTI TRACKING DATASET

Kitti 追蹤資料集上自我姿勢估計的平均值/Rmse

Methods 方法	Seq 02 序列 02	Seq 04	Seq 09	Seq 13	Seq 14
LOAM* [12]	1.184/1.201	2.077/2.771	4.151/5.654	0.641/0.699	0.204/0.265
LOAM [12]	1.135/1.150	1.617/2.023	4.478/6.107	0.620/0.678	0.199/0.261
LeGO-LOAM* [1]	1.096/1.108	1.498/1.831	2.576/3.122	0.668/0.689	0.176/0.216
LeGO-LOAM [1]	1.148/1.161	1.134/1.408	3.037/3.376	0.675/0.697	0.159/0.202
Conf SLAMMOT* (Ours) Conf SLAMMOT* (我們的)	0.919/0.923	0.817/0.941	1.901/2.288	0.606/0.627	0.149/0.185
Conf SLAMMOT (Ours) Conf SLAMMOT (我們的)	0.917/0.925	0.778/0.864	2.441/2.846	0.601/0.632	0.144/0.179

C. Multi-object Tracking Evaluation

C. 多目標追蹤評估

We conducted several experiments on the KITTI Tracking dataset to evaluate the tracking performance of several comparative methods with different detectors, as shown in Table II. The evaluation considers Multi-Object Tracking Accuracy (MOTA), Multi-Object Tracking Precision (MOTP), Recall and Precision based on intersection over union (IoU) thresholds of 0.5. The compared detectors include SECOND [34], PointRCNN [45] and PV-RCNN [32], which are the three common and popular detectors with different detection quality. PV-RCNN combines the advantages of voxel-based and pointnet-based networks, reaching superior performance on various detection datasets compared to the former two [11], [32]. And the compared trackers involve AB3DMOT [22], FG-3DMOT [24], PC3T [23], Conf SLAMMOT# and Conf SLAMMOT. It is noted that the standalone multi-object tracking methods are evaluated by the virtual LiDAR odometry of LeGO-LOAM to provide ego-vehicle pose. Compared to Conf SLAMMOT, the Conf SLAMMOT# method considers only the confidence of object predictions in the data association of tracking, without taking into account the confidence scores.

我們在 KITTI Tracking 資料集上進行了幾個實驗，以評估幾種比較方法與不同偵測器的追蹤效能，如表二所示。評估考慮了多目標追蹤準確度(MOTA)、多目標追蹤精準度(MOTP)、召回率(Recall)與精確度(Precision)，以交集超過結合(IoU)閾值 0.5 為基礎。比較的偵測器包括 SECOND [34]、PointRCNN [45] 與 PV-RCNN [32]，這三種常見且流行的偵測器具有不同的偵測品質。PV-RCNN 結合了 voxel-based 網路與 pointnet-based 網路的優點，在各種偵測資料集上的表現都比前兩者優異[11]、[32]。而比較的追蹤器有 AB3DMOT [22]、FG-3DMOT [24]、PC3T [23]、Conf SLAMMOT# 與 Conf SLAMMOT。值得注意的是，獨立的多目標追蹤方法是透過 LeGO-LOAM 的虛擬 LiDAR 測距來提供自我車輛姿勢。相較於 Conf SLAMMOT，Conf SLAMMOT# 方法在追蹤的資料關聯中只考慮物體預測的置信度，而不考慮置信度分數。

of object detection results. That is, the covariance matrix of the detection results during association is obtained using the overall uncertainty of the detector determined by the sensor's error characteristics, just like in the FG-3DMOT method. From the table, it is evident that the tracking results of the PV-RCNN-based methods generally surpass those of SECOND and PointRCNN. This indirectly confirms that higher detection performance correlates with better tracking effects. Compared to Conf SLAMMOT#, considering the confidence scores of the detection result bounding boxes in Conf SLAMMOT will also help to adaptively adjust the implicit object association range, thereby enhancing tracking performance. From the comparative results of various tracking methods, it can be seen that the Conf SLAMMOT has achieved competitive results. The reason is that its confidence-guided implicit data association integrates the advantages of several methods, and the accuracy of simultaneously optimized ego-pose estimation by joint graph optimization backend is higher than when only using LiDAR odometry as input, which also enhances its tracking performance.

的不確定性。也就是說，關聯過程中偵測結果的協方差矩陣是利用感測器誤差特性所決定的偵測器整體不確定性來取得，就像 FG-3DMOT 方法一樣。從表中可以看出，基於 PV-RCNN 方法的追蹤結果普遍超越 SECOND 和 PointRCNN。這間接印證了更高的偵測效能與更好的追蹤效果相關。相較於 Conf SLAMMOT#，在 Conf SLAMMOT 中考量偵測結果邊界框的置信度分數，也有助於適應性調整隱含物件關聯範圍，進而提升追蹤效能。從各種追蹤方法的比較結果可以看出，Conf SLAMMOT 取得了具有競爭力的結果。究其原因，是其信心導引的隱含資料關聯整合了多種方法的優點，且透過聯合圖表最佳化後端同步最佳化的自我姿勢估計精確度，比僅使用 LiDAR 測距為輸入時更高，也提升了其追蹤效能。

TABLE II 表二
EVALUATION RESULT ON KITTI TRACKING DATASET USING DIFFERENT DETECTORS AND TRACKING

Detector 偵測器	Tracker 追蹤器	MOTA	MOTP	Recall 召回	Precision 精確度
SECOND 34	AB3DMOT 22	0.750	0.861	0.791	0.903
	FG-3DMOT 24	0.779	0.872	0.859	0.915
	PC3T 23; PC3T 23 ;	0.789	0.870	0.867	0.918
	Conf SLAMMOT \#	0.796	0.873	0.862	0.941
	Conf SLAMMOT	0.804	0.873	0.864	0.936
PointRCNN [45]	AB3DMOT 22	0.809	0.865	0.847	0.904
	FG-3DMOT 24.	0.828	0.879	0.889	0.937
	PC3T 23; PC3T 23 ;	0.831	0.881	0.895	0.934
	Conf SLAMMOT ^P	0.833	0.802	0.893	0.946
	Conf SLAMMOT	0.848	0.881	0.901	0.946
PV-RCNN 32	AB3DMOT 22.	0.822	0.872	0.878	0.921
	FG-3DMOT 24	0.837	0.882	0.886	0.949
	PC3T 23; PC3T 23 ;	0.847	0.880	0.899	0.945
	Conf SLAMMOT \#	0.839	0.876	0.879	0.950
	Conf SLAMMOT	0.852	0.882	0.898	0.951

D. Overall Performance Evaluation

D. 整體績效評估

We also evaluate the overall performance of Conf SLAMMOT by assessing the accuracy of state estimation for moving objects with several open-source baseline SLAMMOT methods. Specifically, we compare the multi-object tracking performance in terms of RMSE for longitudinal (long) and lateral (lat) position and yaw angle estimation, as shown in Table III. These methods include LIO-SEGMOT [8] and DL-SLOT [9], both of which utilize LeGO-LOAM for LiDAR odometry and employ PV-RCNN for object detection to ensure consistency in comparison. Herein, several objects with different IDs are selected for error analysis. It can be seen that Conf SLAMMOT shows more advantages compared to other SLAMMOT methods. For example, the tracked frame length of object 2 in sequence 04, other methods would occur ID switching. This is because it considers the confidence of prediction and detection during tracking, which is beneficial for better data association,

我們也以幾種開放原始碼的 SLAMMOT 方法來評估移動物件的狀態估計準確度，藉此評估 Conf SLAMMOT 的整體效能。這些方法包括 LIO-SEGMOT [8] 和 DL-SLOT [9]，這兩種方法都使用 LeGO-LOAM 進行 LiDAR 測距，並使用 PV-RCNN 進行物件偵測，以確保比較結果的一致性。在此，選取數個不同 ID 的物件進行誤差分析。可以看出，與其他 SLAMMOT 方法相比，Conf SLAMMOT 顯示出更多優勢。舉例來說，序列 04 中物件 2 的追蹤幀長，其他方法會發生 ID 切換。這是因為它在追蹤過程中考量了預測與偵測的信賴度，有利於更好的資料關聯。

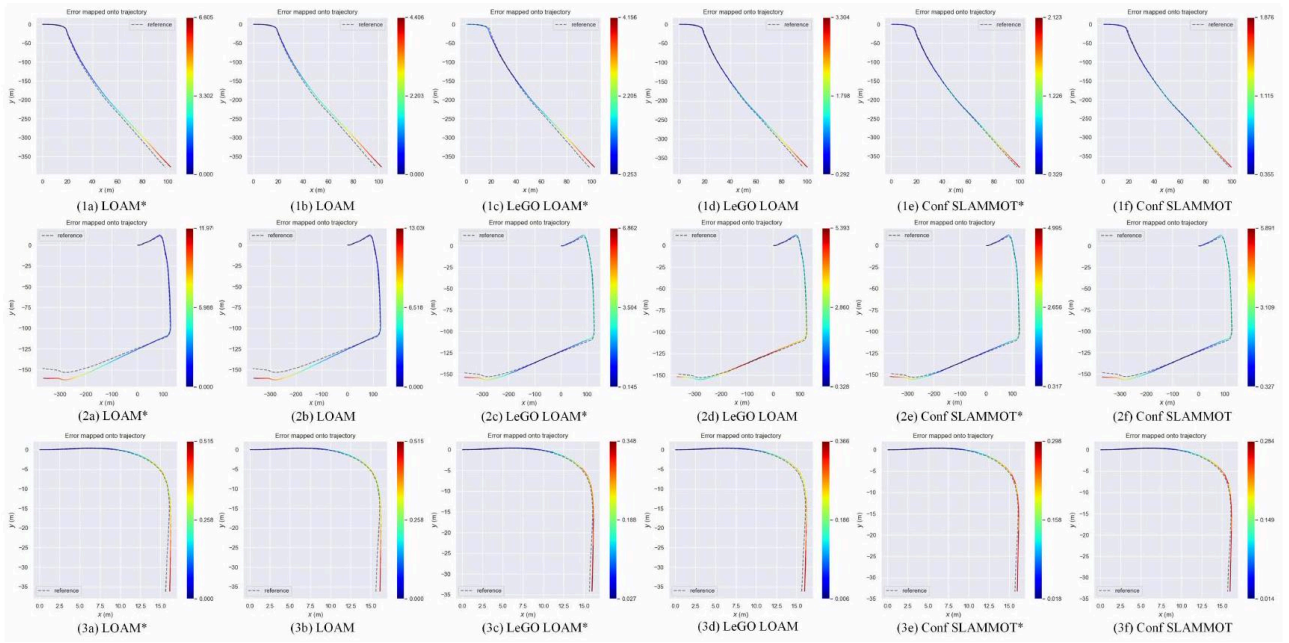


Fig. 4. Ego-trajectory error maps for different methods in KITTI Tracking dataset. The results for sequence 04, 09, 14 are shown in the first three rows. The gray dashed line denotes the ground truth.

圖 4. KITTI Tracking 資料集中不同方法的自我軌跡誤差圖。前三行顯示序列 04, 09, 14 的結果。灰色虛線表示地面真值。

especially in challenging scenarios that there are continuous missed detections. The improvement in tracking results will further enhance ego-vehicle state estimation and achieve an overall performance boost through joint graph optimization.

特別是在持續漏檢的挑戰性場景中。追蹤結果的改善將進一步強化自我車輛的狀態估計，並透過聯合圖形最佳化達到整體效能的提升。

Furthermore, Fig. 5 shows the visualization results of LIOSEGMOT and the proposed Conf SLAMMOT method on different sequences. We tested the aforementioned compared methods on multiple sequences. Due to space limitations, we only present the results of two compared methods on representative sequences here. Additionally, considering that both LIO-SEGMOT and our proposed Conf SLAMMOT utilize factor graph-based tracking methods, the effectiveness of the confidence-guided implicit data association is more intuitively demonstrated through the figures. Three sequences (00, 02, 03) are presented, with (a-d) and (a*-d*) representing different moments of LIO-SEGMOT and Conf SLAMMOT runs. In sequence 00, object ID 1 is heavily occluded by the adjacent object for many consecutive frames, as shown in (1b – 1c) and (1b* – 1c*). Due to the accumulation of errors in the predicted state, ID switches can easily occur when recovering tracking, as seen in the result of LIO-SEGMOT in (1d). However, even with significant temporary occlusions, Conf SLAMMOT correctly tracked the object by successfully associating the currently detected state with the predicted state, as shown in (1d*). In sequence 02, object ID 0, as shown in (2b – 2c) and (2b* – 2c*), as well as in sequence 03 for object IDs 14, as shown in (3b-3c), and object ID 11, as shown in (3b* – 3c*), consecutive missed detections occurred due to the objects being temporarily occluded and being far away. Compared to the ID switches that occurred in LIOSEGMOT after recovery, as shown in (2d) and (3d), the proposed method was able to correctly track these objects, as shown in (2d*) and (3d*). These visualization results further demonstrate the effectiveness of Conf SLAMMOT. All kinds of experimental results demonstrate the superior performance

此外，圖 5 顯示 LIOSEGMOT 與所提出的 Conf SLAMMOT 方法在不同序列上的可視化結果。我們在多個序列上測試了上述的比較方法。由於篇幅所限，我們在此只列出兩種比較方法在代表性序列上的結果。此外，考慮到 LIO-SEGMOT 和我們提出的 Conf SLAMMOT 都使用了基於因子圖的追蹤方法，透過圖表可以更直覺地展示置信導向隱含資料關聯的有效性。下面展示三個序列 (00, 02, 03)，其中 (a-d) 和 (a*-d*) 代表 LIO-SEGMOT 和 Conf SLAMMOT 運行的不同時刻。在序列 00 中，如 (1b-1c) 和 (1b*-1c*) 所示，物件 ID 1 在許多連續畫面中被鄰近物件嚴重遮蔽。由於預測狀態的錯誤累積，在恢復追蹤時很容易發生 ID 切換，如 (1d) 中 LIO-SEGMOT 的結果所示。然而，即使有明顯的暫時遮蔽，Conf SLAMMOT 也能透過成功地將目前偵測到的狀態與預測狀態聯繫起來，正確地追蹤到物體，如 (1d*) 所示。在序列 02 中，如 (2b-2c) 和 (2b*-2c*) 所示的對象 ID 0，以及序列 03 中如 (3b-3c) 所示的對象 ID 14 和 (3b*-3c*) 所示的對象 ID 11，由於對象被暫時遮擋且距離較遠，發生了連續的漏檢。與 LIOSEGMOT 在復原後發生的 ID 切換相比，如 (2d) 和 (3d) 所示，建議的方法能夠正確追蹤這些物件，如 (2d*) 和 (3d*) 所示。這些可視化結果進一步證明了 Conf SLAMMOT 的有效性。各種實驗結果都證明了其優異的性能

of Conf SLAMMOT. It not only addresses the limitations in challenged scenes compared to other baseline SLAMMOT but also improves ego-vehicle pose estimation and multi-object tracking compared to standalone SLAM and MOT methods.

的 Conf SLAMMOT。與其他基準 SLAMMOT 相比，它不僅解決了挑戰性場景的限制，而且與獨立的 SLAM 和 MOT 方法相比，還改善了自我車輛姿勢估計和多目標追蹤。

TABLE III 表三
EVALUATION OF MULTI-OBJECT TRACKING FOR DIFFERENT METHODS IN KitTI Tracking Dataset

在 KitTI 追蹤資料集中，評估不同方法的多目標追蹤效果

Methods 方法	Seq/Object ID 序列/物件 ID	Tracked Frame Length 追蹤框架長度	RMSE (long/m)	RMSE (lat/m)	RMSE (rad)
LIO-SEGMOT 8	04/2	85	0.786	0.791	0.233
	09/15	97	0.339	0.982	0.217
	09/19	46	1.350	0.648	0.364
	11/2	153	1.232	0.512	0.281
DL-SLOT 9	04/2	89	0.679	0.810	0.278
	09/15	97	0.396	0.728	0.188
	09/19	46	1.301	0.656	0.422
	11/2	154	1.281	0.330	0.284
Conf SLAMMOT (Ours) Conf SLAMMOT (我們的)	04/2	230	0.653	1.279	0.239
	09/15	97	0.321	0.778	0.181
	09/19	46	1.249	0.656	0.308
	11/2	157	1.223	0.290	0.277

E. Implementation Efficiency Evaluation

E.執行效率評估

All of the experiments are implemented on a laptop with an Intel Core i7-10750H 2.60 GHz CPU and 12 GB RAM. And we compute the average time consumption of the main functional modules except for object detection, which can be accelerated by GPU. Specifically, the average time consumed for LiDAR odometry to process one scan is approximately 7 ms . The average time consumption for graph optimization backend (integrating the confidence-guided implicit data association) is approximately 62.4 ms in the experiments. The

proposed Conf SLAMMOT can be implemented in real-time.

所有實驗都是在搭載 Intel Core i7-10750H 2.60 GHz CPU 和 12 GB RAM 的筆記型電腦上實作。而我們計算的是主要功能模組的平均耗時，除了物體偵測之外，因為物體偵測可以使用 GPU 加速。具體而言，LiDAR 測距處理一次掃描的平均耗時約為 7 ms。在實驗中，圖形最佳化後端（整合信心導引的隱含資料關聯）的平均耗時約為 62.4 ms。所提出的 Conf SLAMMOT 可以即時執行。

V. ConCluSION V.ConCluSION

This paper proposes a Conf SLAMMOT system that tightly couples the LiDAR SLAM and confidence-guided implicit

本論文提出一種 Conf SLAMMOT 系統，可將 LiDAR SLAM 與信心導引的隱式

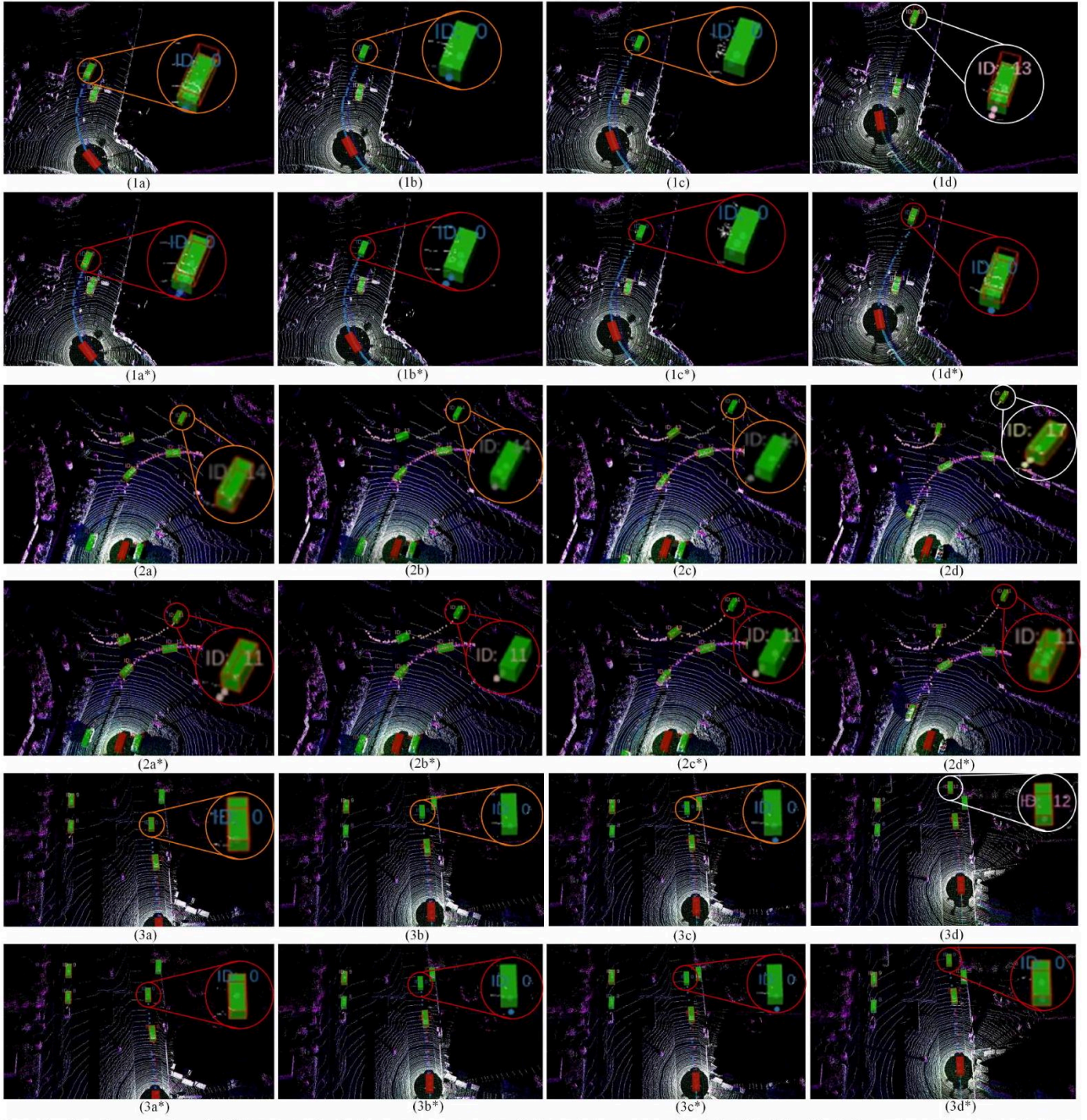


Fig. 5. The visualization results of different methods in continuous missed detection scenes. (1a – 1d), (2a – 2d), (3a – 3d) represent the visualization results of LIO-SEGMOT at consecutive moments on sequences 00, 02, and 03, respectively; (1a* – 1d*), (2a* – 2d*), (3a* – 3d*) represent the visualization results of the proposed Conf SLAMMOT at consecutive moments on sequences 00, 02, and 03, respectively. Red and green fully covered bounding boxes denote the ego-vehicle and tracked objects with different IDs, and red rectangular border denotes detected objects. Due to temporary occlusion or distance of objects, there are continuous frame missed detection; these objects are highlighted and enlarged with circles

for display. For example, object ID o in $(1b^* - 1c^*)$, object ID 11 in $(2b^* - 2c^*)$, and object ID o in $(3b^* - 3c^*)$, and as shown in $(1d^*)$, $(2d^*)$, and $(3d^*)$, Conf SLAMMOT can accurately resume tracking without ID switching. In sequence $o2$, the reason for the differing initial object IDs between the two methods is that the vehicle has already traveled a longer distance, with more moving objects and random ID assignments. In contrast, sequences $o0$ and $o3$ represent scenarios where the vehicle has just started moving.

圖 5.不同方法在連續漏檢場景中的可視化結果。 $(1a - 1d)$, $(2a - 2d)$, $(3a - 3d)$ 分別表示 LIO-SEGMOT 在序列 $o0$ 、 $o2$ 、 $o3$ 的連續時刻的可視化結果； $(1a^* - 1d^*)$, $(2a^* - 2d^*)$, $(3a^* - 3d^*)$ 分別表示所提出的 Conf SLAMMOT 在序列 $o0$ 、 $o2$ 、 $o3$ 的連續時刻的可視化結果。紅色與綠色完全覆蓋的邊界框分別代表不同 ID 的自我車輛與追蹤物件，紅色矩形邊界代表偵測到的物件。由於物體暫時被遮蔽或距離較遠，因此有連續漏測的畫面；這些物體會以圓形高亮顯示並放大顯示。例如， $(1b^* - 1c^*)$ 中的物件 ID 為 o ， $(2b^* - 2c^*)$ 中的物件 ID 為 11 ， $(3b^* - 3c^*)$ 中的物件 ID 為 o ，以及 $(1d^*)$, $(2d^*)$ 和 $(3d^*)$ 中的物件 ID 為 o ，Conf SLAMMOT 可以在不切換 ID 的情況下精確地恢復追蹤。在序列 $o2$ 中，兩種方法的初始物件 ID 不同的原因是，車輛已經行駛了較長的距離，有較多的移動物件和隨機 ID 分配。相比之下，序列 $o0$ 和序列 $o3$ 代表車輛剛開始移動的情況。

data association based multi-object tracking. This method can not only optimize the estimated state of both the ego vehicle and moving objects simultaneously, but also effectively recovers tracking of continuously missed detections based on confidence-guided data association, leading to a stable backend and improved estimation performance. Comparing Conf SLAMMOT with other baseline methods, various experiments demonstrate that the presented method achieves competitive accuracy and performance in terms of ego-pose estimation and object state estimation. In addition, our experiments demonstrate the superior performance of Conf SLAMMOT in scenarios where there are some missed detections due to

基於資料關聯的多物體追蹤。此方法不僅能同時優化自我車輛與移動物體的估計狀態，同時也能在信心導向資料關聯的基礎上有效地恢復連續遺漏偵測的追蹤，從而獲得穩定的後端並提高估計性能。將 Conf SLAMMOT 與其他基準方法比較，各種實驗證明所提出的方法在自我姿態估計與物體狀態估計方面達到具有競爭力的精確度與效能。此外，我們的實驗也證明了 Conf SLAMMOT 在因下列原因而有一些遺漏偵測的情況下，仍有優異的表現。

objects being either occluded or distant, which is meaningful in real-world applications.

物件不是被遮蔽就是距離很遠，這在現實世界的應用中很有意義。

Moreover, in this work, we neglect objects or the surrounding environment with different categories, which may be detrimental not only to the proposed Conf SLAMMOT method but also to related object perception methods. Therefore, incorporating semantic information will be a direction for future work. Additionally, integrating multiple motion models into multi-object tracking for dealing with moving objects with ambiguous motion status and irregular motion patterns may be a promising extension of research works.

此外，在本工作中，我們忽略了具有不同類別的物件或周遭環境，這可能不僅不利於所提出的 Conf SLAMMOT 方法，也不利於相關的物件感知方法。因此，結合語意資訊將是未來的工作方向。此外，將多個運動模型整合至多目標追蹤中，以處理具有模糊運動狀態與不規則運動模式的移動物體，也可能是一個很有希望的研究工作延伸。

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